

论文写作 Agent 测试文档

Abstract

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This study explores the application of large language models in automated academic paper writing by constructing a dedicated paper-writing intelligent agent (Agent). This Agent provides comprehensive assistance throughout the entire writing process, encompassing literature retrieval, outline generation, paragraph composition, and formatting standardization. Experimental results demonstrate that the proposed Agent significantly enhances the efficiency of academic paper writing while maintaining the规范性 (normativity) and logical coherence of scholarly expression. Additionally, a Markdown template suitable for testing the paper-writing Agent, along with prompts for directly generating images, is provided for practical application.

Keywords: 大语言模型, 智能体, 论文写作, 学术辅助, 自然语言处理

2. 引言

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With the rapid advancement of large language model technologies, artificial intelligence has demonstrated significant potential in the field of content generation. Academic paper writing is characterized by its rigorous structure, standardized language, and stringent logical coherence, which traditionally demands substantial time and effort. Consequently, the development of a dedicated academic paper writing agent holds substantial practical value for enhancing efficiency and quality in scholarly composition.

The primary contributions of this study are outlined as follows: First, we propose an architecture for an academic paper writing agent specifically tailored to scholarly contexts. Second, we implement modular functionalities including outline generation, literature citation management, and mathematical formula formatting. Third, we

validate the system's effectiveness and usability through multiple sets of comparative experiments, with detailed results presented in subsequent sections.

3. 相关工作

3.1 基于大模型的文本生成

3.1 Text Generation Based on Large Language Models

In recent years, significant breakthroughs have been achieved in long-text generation through the development of large language models such as GPT and LLaMA. Consequently, numerous studies have explored the application of these models to the field of scientific and technical document writing, aiming to leverage their advanced capabilities for enhancing the efficiency and quality of such documentation.

3.2 智能体 (Agent) 在专业领域的应用

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Intelligent agents, through their capabilities of tool invocation, planning, and reflection, demonstrate a remarkable capacity to accomplish complex tasks. These agents have been extensively studied and applied across various professional domains, including education, healthcare, and code development, where their functionalities have been leveraged to enhance operational efficiency and problem-solving capabilities.

4. 系统设计

4.1 整体架构

4.1 Overall Architecture

The academic paper writing agent primarily comprises the following functional modules: the user intention analysis module, which is responsible for interpreting and clarifying user requirements; the literature retrieval and summarization module, which conducts targeted searches of relevant academic resources and generates concise

summaries; the paper outline planning module, which systematically organizes the structural framework of the manuscript; the paragraph automatic generation module, which produces coherent and contextually appropriate textual content based on predefined parameters; and the format verification and refinement module, which ensures adherence to academic writing standards through comprehensive style checks and aesthetic enhancements.

4.2 工作流程

4.2 Workflow

The operational procedure of the system unfolds through the following sequential stages: First, the user initiates the process by submitting the paper's thematic focus, specific requirements, and designated word count through the interface. Subsequently, the intelligent agent autonomously generates a hierarchical three-level outline that systematically organizes the paper's structural framework. Following outline approval, the system proceeds to compose the main text section by section, incorporating relevant citations and mathematical formulations as required by the academic discipline. Finally, the completed manuscript is exported in standardized document formats, including Markdown or Microsoft Word, ensuring compatibility with conventional academic submission protocols.

5. 实验与结果

5.1 实验设置

5.1 Experimental Setup

The foundational models employed in this study are LLaMA-3 and GPT-3.5, which serve as the primary language model architectures for comparison. The experimental task is designed to evaluate the models' performance in generating a computer science academic paper of approximately 3,000 words in length. To assess the quality and efficiency of the generated outputs, four key evaluation metrics are adopted: (1) writing

duration, which measures the time required for each model to complete the paper generation task; (2) structural integrity, which evaluates the coherence and logical organization of the generated content; (3) linguistic fluency, which assesses the grammatical correctness and naturalness of the language; and (4) formatting compliance, which examines the adherence of the generated paper to standard academic formatting conventions.

5.2 结果分析

5.2 Results Analysis

The experimental results demonstrate that the paper-writing agent significantly outperforms manual writing in terms of efficiency while simultaneously maintaining a high level of academic quality.

Table 1

指标	人工写作	论文 Agent
平均耗时	3 天	45 分钟
结构完整度	82%	89%
语言通顺度	85%	92%
格式规范率	78%	95%

6. 结论与展望

6. 结论与展望

This study has successfully designed and implemented an intelligent agent tailored for academic paper writing, which is capable of efficiently providing comprehensive assistance throughout the entire process, including the generation of paper outlines, composition of main text, and formatting adjustments. Looking ahead, future research endeavors could focus on the integration of multi-agent collaboration frameworks to enhance the system's capacity for handling complex writing tasks through coordinated efforts among specialized agents. Additionally, the incorporation of real-time literature update mechanisms would ensure that the generated content remains current and

aligned with the latest research developments. Furthermore, the establishment of a feedback loop involving domain experts could significantly improve the professionalism and reliability of the system's outputs, thereby making it a more valuable tool for academic writing.

可直接生成的图片（用于测试插图）

Images That Can Be Directly Generated (For Testing Illustrations)

To generate the relevant images, one can simply input the following prompts into an image - generation tool.

图1：论文写作 Agent 系统架构图

Figure 1: Architectural Diagram of the Academic Paper Writing Agent System

The diagram illustrates the system architecture of an intelligent agent designed for academic paper writing, adopting a minimalist technological aesthetic with a dominant blue color scheme. The structure is organized into four hierarchical layers: the User Layer, Planning Layer, Execution Layer, and Tool Layer, each represented as distinct modular components. The flowchart-style visualization employs clean, simplified lines on a white background to ensure clarity and suitability for academic publication. This modular design facilitates a transparent depiction of interactions between layers, with directional connections indicating the workflow sequence from user input to tool-assisted execution.

图2：实验对比柱状图

Figure 2 presents a comparative bar chart illustrating the performance differences between human-authored academic writing and AI Agent-generated content across four key evaluation metrics: time consumption, structural integrity, linguistic fluency, and formatting compliance. The visualization employs an academic color scheme with subdued, professional tones to ensure clarity and visual neutrality, set against a clean white background for optimal readability. Each metric is represented by paired bars enabling direct comparison between human and AI-generated outputs, with precise

numerical values or percentage rates (where applicable) annotated to facilitate quantitative analysis. The chart's minimalist design adheres to standard academic conventions, eliminating extraneous elements while maintaining sufficient visual distinction between data categories through controlled use of color and bar width.

图3：智能体工作流程图

****Figure 3: Workflow Diagram of the Intelligent Agent****

This diagram illustrates the academic workflow of the intelligent agent, structured as a unidirectional process encompassing five sequential stages: user input acquisition, outline generation, paragraph composition, formatting optimization, and final output delivery. The visual representation employs minimalist technological line art with a blue color scheme, ensuring clarity and alignment with academic illustration standards for scholarly publications.